INCH-POUND

AN846 Rev 8
w/AMENDMENT 1
5 November 2012
SUPERSEDING
AN846 Rev 8
7 June 2011

DETAIL SPECIFICATION SHEET

ELBOW, HOSE, 45°

Reinstated after 7 June 2011. Inactive for new design.
For new design, use SAE-AS5187.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and SAE-AS4843.

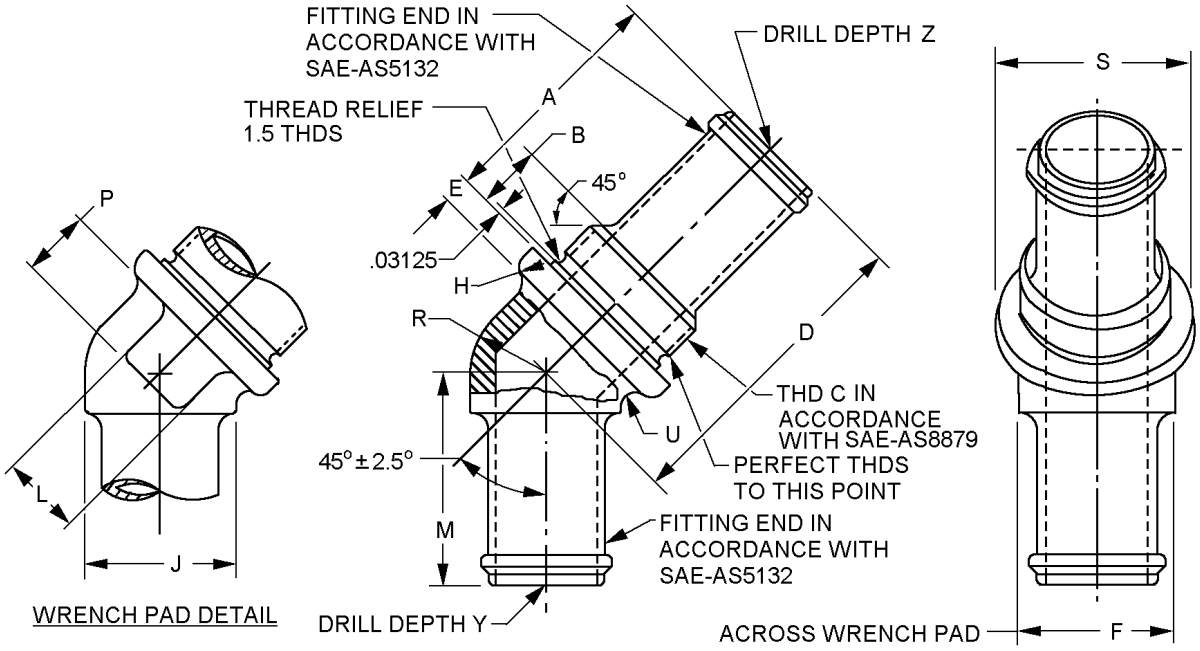


FIGURE 1. 45° elbow dimensions and configuration.

AMSC N/A

FSC 4730

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Dash number	Hose ID inches (mm)	A inches (mm)	B inches (mm)	Thread C Ref SAE-AS8879	D .047 (1.19) -.000 inches (mm)
4	.250 (6.35)	1.938 (49.23)	.438 (11.13)	.4375-20UNJF-3A	2.266 (57.56)
6	.375 (9.53)	1.969 (50.01)	.469 (11.91)	.5625-18UNJF-3A	2.406 (61.11)
8	.500 (12.70)	2.000 (50.80)	.500 (12.70)	.7500-16UNJF-3A	2.500 (63.50)
10	.675 (17.15)	2.031 (51.59)	.531 (13.49)	1.875 -12UNJ-3A	2.641 (67.08)
12	.750 (19.05)	2.063 (52.40)	.563 (14.30)	1.0625 -12UNJ-3A	2.703 (68.66)
16	1.000 (25.40)	2.063 (52.40)	.563 (14.30)	1.3125-12UNJ-3A	2.750 (69.85)
20	1.250 (31.75)	2.063 (52.40)	.563 (14.30)	1.6250-12UNJ-3A	2.938 (74.63)
24	1.500 (38.10)	2.063 (52.40)	.563 (14.30)	1.8750-12UNJ-3A	3.000 (76.20)

Dash number	E inches (mm)	F inches (mm)	H Radius inches (mm)	J Diameter inches (mm)	L Approx. inches (mm)	M +.047 (1.19) -.000 inches (mm)
4	.156 (3.96)	.438 (11.13)	.063 (1.60)	.438 (11.13)	.250 (6.35)	1.609 (40.87)
6	.188 (4.78)	.563 (14.30)	.063 (1.60)	.563 (14.30)	.375 (9.53)	1.656 (42.06)
8	.219 (5.56)	.750 (19.05)	.094 (2.39)	.750 (19.05)	.438 (11.13)	1.688 (42.88)
10	.250 (6.35)	.875 (22.23)	.094 (2.39)	.875 (22.23)	.500 (12.70)	1.734 (44.04)
12	.250 (6.35)	1.063 (27.00)	.094 (2.39)	1.063 (27.00)	.563 (14.30)	1.766 (44.86)
16	.250 (6.35)	1.313 (33.35)	.125 (3.18)	1.313 (33.35)	.688 (17.48)	1.828 (46.43)
20	.281 (7.14)	1.625 (41.28)	.125 (3.18)	1.625 (41.28)	1.000 (25.40)	1.922 (48.82)
24	.281 (7.14)	1.875 (47.63)	.125 (3.18)	1.875 (47.63)	1.125 (28.58)	1.969 (50.01)

Dash number	P Approx inches (mm)	R Radius inches (mm)	S Diameter inches (mm)	U Radius inches (mm)	Y +.047 (1.19) -.000 inches (mm)	Z +.047 (1.19) -.000 inches (mm)
4	.219 (5.56)	.219 (5.56)	.625 (15.88)	.063 (1.60)	1.641 (41.68)	2.297 (58.34)
6	.281 (7.14)	.281 (7.14)	.813 (20.65)	.094 (2.39)	1.719 (43.66)	2.469 (62.71)
8	.344 (8.75)	.375 (9.53)	1.000 (25.40)	.094 (2.39)	1.766 (44.86)	2.578 (65.48)
10	.438 (11.13)	.438 (11.13)	1.250 (31.75)	.125 (3.18)	1.848 (46.94)	2.750 (69.85)
12	.500 (12.70)	.531 (13.49)	1.438 (36.53)	.125 (3.18)	1.891 (48.03)	2.828 (71.83)
16	.563 (14.30)	.656 (16.66)	1.688 (42.88)	.125 (3.18)	2.016 (51.21)	2.938 (74.63)
20	.688 (17.48)	.813 (20.65)	2.063 (52.40)	.188 (4.48)	2.141 (54.38)	3.156 (80.16)
24	.750 (19.05)	.938 (22.83)	2.375 (60.33)	.188 (4.48)	2.250 (57.15)	3.281 (83.34)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ±.016 inch (0.41 mm).
4. Break sharp edges .and remove all hanging burrs and slivers
5. Machined surfaces shall be finished to 125µ-in in Ra, forged surfaces shall be 250µ-in Ra, unless otherwise specified on the figures. Surface finish shall be in accordance with ASME B46.1.
6. For design features purposes, this standard takes precedence over documents referenced herein.
7. Referenced documents shall be of the issue in effect on date of invitation for bid.

FIGURE 1. 45° elbow dimensions and configuration - Continued.

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REQUIREMENTS:

Dimensions and configuration shall be in accordance with figure 1.

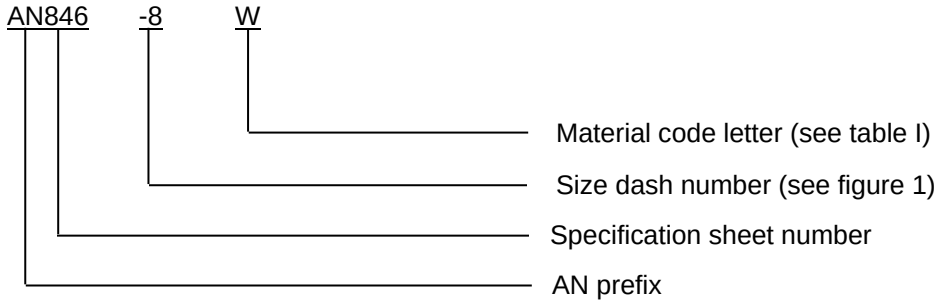
Materials and finishes shall be in accordance with SAE-AS4843/2, see table I for material code.

TABLE I. Material and code letters.

Code letter	Material
Blank	Copper alloy
D 1/	Aluminum alloy 2014-T6 or 2024-T6 or T851
J	Corrosion resistant steel (CRES), type 304
K	CRES, type 316
S	CRES, type 347
R	CRES. Type 321
T 2/	Titanium alloy
W	Aluminum alloy 7075-T73

1/ Inactive for new design, for new design use aluminum alloy 7075 to improve stress corrosion resistance and tensile strength.
2/ Not for use in oxygen systems.

Part or Identifying Number (PIN): The PIN consists of the letter "AN" the specification sheet number, a dash number for tube sizes, and a code letter for material type. Unassigned PIN's shall not be used.



PIN example: AN846-8WP indicates a 45° elbow, hose to hose, .500 inch (12.70 mm) ID, aluminum alloy 7075-T73.

Marking: Part shall be permanently marked with the AN PIN, and include the manufacturers CAGE, name, or trademark.

Table II provides a detailed cross-reference of AN846 PINs and replacement SAE-AS5187 PINs. Users are cautioned to evaluate replacements for their particular application.

CAUTION: The superseding information is valid as of the date of this specification and may be superseded by subsequent revisions of the superseding document.

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TABLE II. Cross-reference data. 1/

AN PIN (inactive for new design)	Hose Size	Tube Size	For new design AS PIN
AN846-4	0.250	0.250	AS5187B04
AN846-4D	0.250	0.250	AS5187W04
AN846-4J	0.250	0.250	AS5187J04
AN846-4K	0.250	0.250	AS5187K04
AN844-4R	0.250	0.250	AS5187R04
AN844-4S	0.250	0.250	AS5187R04
AN846-4T	0.250	0.250	None
AN846-4W	0.250	0.250	AS5187W04
AN846-6	0.375	0.375	AS5187B06
AN846-6D	0.375	0.375	AS5187W06
AN846-4J	0.375	0.375	AS5187J06
AN846-4K	0.375	0.375	AS5187K06
AN846-6R	0.375	0.375	AS5187R06
AN846-6S	0.375	0.375	AS5187R06
AN846-4T	0.375	0.375	None
AN846-6W	0.375	0.375	AS5187W06
AN846-8	0.500	0.500	AS5187B08
AN846-8D	0.500	0.500	AS5187W08
AN846-8J	0.500	0.500	AS5187J08
AN846-8K	0.500	0.500	AS5187K08
AN846-8R	0.500	0.500	AS5187R08
AN846-8S	0.500	0.500	AS5187R08
AN846-8T	0.500	0.500	None
AN846-8W	0.500	0.500	AS5187W08
AN846-10	0.625	0.625	AS5187B10
AN846-10D	0.625	0.625	AS5187W10
AN846-10J	0.625	0.625	AS5187J10
AN846-10K	0.625	0.625	AS5187K10
AN846-10R	0.625	0.625	AS5187R10
AN846-10S	0.625	0.625	AS5187R10
AN846-10T	0.625	0.625	None
AN846-10W	0.625	0.625	AS5187W10

See note at end of table.

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TABLE II. Cross-reference data - Continued. 1/

AN PIN (inactive for new design)	Hose Size	Tube Size	For new design AS PIN
AN846-12	0.750	0.750	AS5187B12
AN846-12D	0.750	0.750	AS5187W12
AN846-12J	0.750	0.750	AS5187J12
AN846-12K	0.750	0.750	AS5187K12
AN846-12R	0.750	0.750	AS5187R12
AN846-12S	0.750	0.750	AS5187R12
AN846-12T	0.750	0.750	None
AN846-12W	0.750	0.750	AS5187W12
AN846-16	1.000	1.000	AN5187B16
AN846-16D	1.000	1.000	AS5187W16
AN846-16J	1.000	1.000	AS5187J16
AN846-16K	1.000	1.000	AS5187K16
AN846-16R	1.000	1.000	AS5187R16
AN846-16S	1.000	1.000	AS5187R16
AN846-16T	1.000	1.000	None
AN846-16W	1.000	1.000	AS5187W16
AN846-20	1.250	1.250	AS5187B20
AN846-20D	1.250	1.250	AS5187W20
AN846-20J	1.250	1.250	AS5187J20
AN846-20K	1.250	1.250	AS5187K20
AN846-20R	1.250	1.250	AS5187R20
AN846-20S	1.250	1.250	AS5187R20
AN846-20T	1.250	1.250	None
AN846-20W	1.250	1.250	AS5187W20
AN846-24	1.500	1.500	AS5187B24
AN846-24D	1.500	1.500	AS5187W24
AN846-24J	1.500	1.500	AS5187J24
AN846-24K	1.500	1.500	AS5187K24
AN846-24R	1.500	1.500	AS5187R24
AN846-24S	1.500	1.500	AS5187R24
AN846-24T	1.500	1.500	None
AN846-24W	1.500	1.500	AS5187W24

1/ For new design use material designator R and W.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Referenced documents. In addition to SAE-AS4843, this document references the following:

ASME B46.1
SAE-AS4843/2
SAE-AS5132
SAE-AS5187
SAE-AS8879

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2012-083)

Review activity:
Air Force - 71

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